Final project work

Exam 2.

1) $r(t) = 1$ $\quad \frac{C(s)}{R(s)} = \frac{W_n^2}{s^2 + 2\zeta W_n s + W_n^2}$

$W_n = a + b = 2 + 10 = 12$ $\quad \zeta = \frac{b}{2c} = \frac{10}{2 \cdot 12} = 0.4166$

a)

1a) $\boxed{W_n = 12}$ $\quad$ 2a) $\boxed{\zeta = 0.4166}$

3a) $W_d = W_n \sqrt{1-\zeta^2} = 12\sqrt{1-0.4166^2} = \boxed{10.91}$

4a) $\beta = \sin^{-1}(W_d/W_n) = \boxed{69.38°} = \boxed{1.141}$

5) $T_p = \pi / W_d = \boxed{0.288}$

6) ~~$T_s = 4/\zeta W_n$~~ $T_s = \frac{4}{\zeta W_n} = \boxed{0.8}$

7) $T_r = \frac{\pi - \beta}{W_d} = \frac{\pi - 1.141}{10.91} = \boxed{0.183}$

8) $M_p = e^{-\pi\zeta/\sqrt{1-\zeta^2}} = e^{-\pi(0.4166)/\sqrt{1-(0.4166)^2}} = \boxed{0.237}$

9) $M_p\% = M_p \cdot 100\% = \boxed{23.7\%}$

(1) 4) 10) $C_{ss} = C(\infty)$

$$\lim_{s \to 0} \frac{144}{s^2 + 24s + 144} = \frac{144}{144} = 1$$